

Stat 517

Homework # 2

1. Let F have an F distribution with parameters r_1 and r_2 . Argue that $1/F$ has an F distribution with parameters r_2 and r_1 .

Sol. The definition of F -distribution is

$$F = \frac{V_1/r_1}{V_2/r_2}$$

where $V_1 \sim \chi^2(r_1)$ and $V_2 \sim \chi^2(r_2)$; V_1 and V_2 are statistically independent. Thus, the reciprocal,

$$1/F = \frac{V_2/r_2}{V_1/r_1},$$

also has F -distribution with parameter r_2 and r_1 .

2. Let $\bar{X}$ be the mean of a random sample of size n from a distribution that is $N(\mu, 9)$. Find n such that $P(\bar{X} - 1 < \mu < \bar{X} + 1) = 0.90$, approximately

Sol. Since $(\bar{X} - \mu) / (3/\sqrt{n}) \sim N(0, 1)$,

$$\begin{aligned} P(\bar{X} - 1 < \mu < \bar{X} + 1) &= P\left(-\frac{\sqrt{n}}{3} < \frac{\sqrt{n}(\bar{X} - \mu)}{3} < \frac{\sqrt{n}}{3}\right) \\ &= 2\Phi\left(\frac{\sqrt{n}}{3}\right) - 1, \end{aligned}$$

where $\Phi(x) = \int_{-\infty}^x (\sqrt{2\pi})^{-1} e^{-t^2/2} dt$. Setting this equal to 0.90 gives

$$2\Phi\left(\frac{\sqrt{n}}{3}\right) - 1 = 0.90 \implies \Phi\left(\frac{\sqrt{n}}{3}\right) = 0.95.$$

Since the cdf is strictly increasing, we may invert it:

$$\frac{\sqrt{n}}{3} = \Phi^{-1}(0.95) = z_{0.05} \approx 1.645.$$

Therefore the required sample size is

$$n = (3\Phi^{-1}(0.95))^2 = (3 \times 1.645)^2 \approx 24.35,$$

so we round up to $n = 25$.

3. Let two independent random samples, each of size 10, from two normal distributions $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ yield $\bar{x} = 4.8$, $s_1^2 = 8.64$, $\bar{y} = 5.6$, $s_2^2 = 7.88$. Find a 95% confidence interval for $\mu_1 - \mu_2$.

Sol. Let $X_1, \dots, X_{10} \sim N(\mu_1, \sigma^2)$ and let $Y_1, \dots, Y_{10} \sim N(\mu_2, \sigma^2)$, with a common (unknown) variance σ^2 . Since the samples are independent,

$$\bar{X} - \bar{Y} \sim N\left(\mu_1 - \mu_2, \frac{\sigma^2}{10} + \frac{\sigma^2}{10}\right) = N\left(\mu_1 - \mu_2, \frac{\sigma^2}{5}\right),$$

so that, after standardizing,

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma \sqrt{\frac{1}{10} + \frac{1}{10}}} \sim N(0, 1).$$

Since σ^2 is unknown, we estimate it with the pooled variance

$$\hat{\sigma}^2 = S_p^2 = \frac{(10-1)S_1^2 + (10-1)S_2^2}{(10-1) + (10-1)} = \frac{9(8.64) + 9(7.88)}{18} = \frac{148.68}{18} = 8.26,$$

so $S_p = \sqrt{8.26} \approx 2.874$. Replacing σ by S_p yields a statistic with a t -distribution on $n_1 + n_2 - 2 = 18$ degrees of freedom:

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{10} + \frac{1}{10}}} = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p / \sqrt{5}} \sim t(18).$$

Therefore

$$P\left(-t_{0.025, 18} < \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{S_p / \sqrt{5}} < t_{0.025, 18}\right) = 0.95, \quad t_{0.025, 18} = 2.101,$$

which gives the confidence interval

$$\left[(\bar{X} - \bar{Y}) - t_{0.025, 18} \frac{S_p}{\sqrt{5}}, (\bar{X} - \bar{Y}) + t_{0.025, 18} \frac{S_p}{\sqrt{5}} \right].$$

Plugging in $\bar{x} - \bar{y} = 4.8 - 5.6 = -0.8$ and $\frac{S_p}{\sqrt{5}} = \frac{2.874}{\sqrt{5}} \approx 1.285$,

$$-0.8 \pm (2.101)(1.285) = -0.8 \pm 2.70 \implies (-3.50, 1.90).$$

Since this 95% confidence interval contains 0, there is no significant difference between μ_1 and μ_2 at the 5% level.

4. Let $X_1, X_2, \dots, X_n, X_{n+1}$ be a random sample of size $n+1$, $n > 1$, from a distribution that is $N(\mu, \sigma^2)$. Let $\bar{X} = \sum_{i=1}^n X_i / n$ and $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)$. Find the constant c so that the statistic $c(\bar{X} - X_{n+1})/S$ has the t -distribution. If $n = 8$, determine k such that $P(\bar{X} - kS < X_9 < \bar{X} + kS) = 0.80$. The observed interval $(\bar{x} - ks, \bar{x} + ks)$ is often called an 80% *prediction interval* for X_9 .

Sol. Note that $\bar{X}$ and S^2 are computed from the first n observations only, while X_{n+1} is a *new, independent* observation from the same $N(\mu, \sigma^2)$ distribution. Because X_{n+1} is independent of the first n observations (and hence of $\bar{X}$),

$$E(\bar{X} - X_{n+1}) = \mu - \mu = 0,$$

$$\text{Var}(\bar{X} - X_{n+1}) = \text{Var}(\bar{X}) + \text{Var}(X_{n+1}) = \frac{\sigma^2}{n} + \sigma^2 = \frac{n+1}{n} \sigma^2.$$

Here the full variance σ^2 of the single new observation is added to the variance σ^2/n of the mean, which is what makes a prediction interval wider than a confidence interval. Standardizing,

$$\frac{\bar{X} - X_{n+1}}{\sigma \sqrt{(n+1)/n}} \sim N(0, 1).$$

Since σ^2 is unknown, we replace σ by S . Using $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$ (independent of the numerator),

$$T = \frac{(\bar{X} - X_{n+1})/\sqrt{(n+1)\sigma^2/n}}{S/\sigma} = \frac{\bar{X} - X_{n+1}}{S\sqrt{(n+1)/n}} \sim t_{n-1},$$

which, written in the requested form $c(\bar{X} - X_{n+1})/S$, gives

$$c = \sqrt{\frac{n}{n+1}}.$$

If $n = 8$, then $X_{n+1} = X_9$, $c = \sqrt{8/9}$, and the degrees of freedom are $n - 1 = 7$ (since S^2 uses the $n = 8$ observations). Hence

$$\begin{aligned} P(\bar{X} - kS < X_9 < \bar{X} + kS) &= 0.80 \Rightarrow P\left(\left|\frac{\bar{X} - X_9}{S}\right| < k\right) = 0.80 \\ &\Rightarrow P\left(\sqrt{\frac{8}{9}} \left|\frac{\bar{X} - X_9}{S}\right| < k\sqrt{\frac{8}{9}}\right) = 0.80 \\ &\Rightarrow k\sqrt{\frac{8}{9}} = t_{0.10, 7} = 1.415 \\ &\Rightarrow k = t_{0.10, 7} \times \sqrt{\frac{9}{8}} = 1.415 \times 1.0607 \approx 1.50. \end{aligned}$$

5. Let X have a pdf of the form $f(x; \theta) = \theta x^{\theta-1}$, $0 < x < 1$, zero elsewhere, where $\theta \in \{\theta : \theta = 1, 2\}$. To test the simple hypothesis $H_0 : \theta = 1$ against the alternative simple hypothesis $H_1 : \theta = 2$, use a random sample X_1, X_2 of size $n = 2$ and define the critical region to be $C = \left\{(x_1, x_2) : \frac{3}{4} \leq x_1 x_2\right\}$. Find the power function of the test.

Sol. The power function is

$$\begin{aligned} P(X_1 X_2 \geq \frac{3}{4}) &= \int_0^1 \left[\int_{\min(\frac{3}{4x_1}, 1)}^1 \theta x_2^{\theta-1} dx_2 \right] \theta x_1^{\theta-1} dx_1 \\ &= \int_0^1 \{1 - [\min(\frac{3}{4x_1}, 1)]^\theta\} \theta x_1^{\theta-1} dx_1 \\ &= \int_{\frac{3}{4}}^1 [1 - (\frac{3}{4})^\theta x_1^{-\theta}] \theta x_1^{\theta-1} dx_1 \\ &= [1 - (\frac{3}{4})^\theta] + \theta (\frac{3}{4})^\theta \log \frac{3}{4} \\ &= \begin{cases} 0.03424 & \text{when } \theta = 1 \\ 0.1139 & \text{when } \theta = 2 \end{cases} \end{aligned}$$

6. Let X have a binomial distribution with the number of trials $n = 10$ and with p either $1/4$ or $1/2$. The simple hypothesis $H_0 : p = 1/2$ is rejected, and the alternative simple hypothesis $H_1 : p = 1/4$ is accepted, if the observed value of X_1 , a random sample of size 1, is less than or equal to 3. Find the significance level and the power of the test.

Sol. The rejection region is $X \leq 3$: we reject H_0 (and accept H_1) when the observed number of successes is small. This is reasonable because a smaller success probability p tends to produce fewer successes, so a small value of X is evidence in favor of $H_1 : p = 1/4$.

Significance level α (Type I error). The significance level is the probability of rejecting H_0 when H_0 is true, i.e. when $p = 1/2$:

$$\begin{aligned}\alpha = P(X \leq 3 \mid p = \tfrac{1}{2}) &= \sum_{x=0}^3 \binom{10}{x} \left(\tfrac{1}{2}\right)^x \left(\tfrac{1}{2}\right)^{10-x} = \frac{1}{2^{10}} \sum_{x=0}^3 \binom{10}{x} \\ &= \frac{\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3}}{1024} = \frac{1 + 10 + 45 + 120}{1024} = \frac{176}{1024} = 0.1719.\end{aligned}$$

When $p = 1/2$ the distribution is centered at 5, so the event $X \leq 3$ is only a small left tail; this is why α is small.

Power of the test. The power is the probability of (correctly) rejecting H_0 when H_1 is true, i.e. when $p = 1/4$:

$$\begin{aligned}\text{power} = P(X \leq 3 \mid p = \tfrac{1}{4}) &= \sum_{x=0}^3 \binom{10}{x} \left(\tfrac{1}{4}\right)^x \left(\tfrac{3}{4}\right)^{10-x} \\ &= 0.0563 + 0.1877 + 0.2816 + 0.2503 = 0.7759,\end{aligned}$$

where the four terms correspond to $x = 0, 1, 2, 3$. When $p = 1/4$ the distribution is centered near 2.5, so the event $X \leq 3$ captures most of the probability; this is why the power is large.

Summary. Both quantities are values of the single power function

$$\gamma(p) = P(\text{Bin}(10, p) \leq 3) = \begin{cases} 0.1719 & \text{when } p = 1/2 \quad (\text{significance level } \alpha), \\ 0.7759 & \text{when } p = 1/4 \quad (\text{power}). \end{cases}$$

The same rejection region $X \leq 3$ yields a small α under H_0 and a large power under H_1 precisely because the two distributions are well separated. (Note the trade-off: tightening the region to $X \leq 2$ would lower α to 0.0547 but also drop the power to about 0.526.)

7. Let X_1, X_2 be a random sample of size $n = 2$ from the distribution having pdf $f(x; \theta) = (1/\theta) e^{-x/\theta}$, $0 < x < \infty$, zero elsewhere. We reject $H_0 : \theta = 2$ and accept $H_1 : \theta = 1$ if the observed values of X_1, X_2 , say x_1, x_2 , are such that

$$\frac{f(x_1; 2) f(x_2; 2)}{f(x_1; 1) f(x_2; 1)} \leq \frac{1}{2}.$$

Here $\Omega = \{\theta : \theta = 1, 2\}$. Find the significance level of the test and the power of the test when H_0 is false.

Sol. Step 1: Simplify the rejection region. Since $f(x; \theta) = \frac{1}{\theta} e^{-x/\theta}$, the numerator (at $\theta = 2$) and denominator (at $\theta = 1$) of the likelihood ratio are

$$f(x_1; 2)f(x_2; 2) = \frac{1}{4}e^{-(x_1+x_2)/2}, \quad f(x_1; 1)f(x_2; 1) = e^{-(x_1+x_2)}.$$

Hence the likelihood ratio is

$$\frac{f(x_1; 2)f(x_2; 2)}{f(x_1; 1)f(x_2; 1)} = \frac{\frac{1}{4}e^{-(x_1+x_2)/2}}{e^{-(x_1+x_2)}} = \frac{1}{4}e^{(x_1+x_2)/2},$$

so the test rejects H_0 when

$$\frac{1}{4}e^{(x_1+x_2)/2} \leq \frac{1}{2} \Rightarrow e^{(x_1+x_2)/2} \leq 2 \Rightarrow x_1 + x_2 \leq \log 4.$$

Intuitively, $\theta = 1$ has the smaller mean, so small observed values (a small sum) point toward H_1 ; the rejection region $x_1 + x_2 \leq \log 4$ reflects this.

Step 2: Distribution of the test statistic. The statistic is $X_1 + X_2$, the sum of two independent exponential random variables with mean θ , which follows a $\text{Gamma}(2, \theta)$ distribution. Its cdf (obtained by the double integral below) is $P(X_1 + X_2 \leq z) = 1 - e^{-z/\theta}(1 + \frac{z}{\theta})$.

Step 3: Power function.

$$\begin{aligned} \gamma(\theta) &= P(X_1 + X_2 \leq \log 4) = \int_0^{\log 4} \left[\int_0^{\log 4 - x_1} \frac{1}{\theta} e^{-x_2/\theta} dx_2 \right] \frac{1}{\theta} e^{-x_1/\theta} dx_1 \\ &= \int_0^{\log 4} \left[1 - e^{-(\log 4 - x_1)/\theta} \right] \frac{1}{\theta} e^{-x_1/\theta} dx_1 \\ &= (1 - 4^{-1/\theta}) - \frac{4^{-1/\theta} \log 4}{\theta}, \end{aligned}$$

where we used $e^{-(\log 4)/\theta} = 4^{-1/\theta}$.

Step 4: Evaluate ($\log 4 \approx 1.3863$).

$$\theta = 2 \text{ (} H_0 \text{ true)} : \quad \gamma(2) = (1 - 4^{-1/2}) - \frac{4^{-1/2} \log 4}{2} = (1 - \frac{1}{2}) - \frac{\frac{1}{2}(1.3863)}{2} = 0.1534,$$

$$\theta = 1 \text{ (} H_1 \text{ true)} : \quad \gamma(1) = (1 - 4^{-1}) - \frac{4^{-1} \log 4}{1} = (1 - \frac{1}{4}) - \frac{1}{4}(1.3863) = 0.4034.$$

Conclusion. Both values come from the single power function

$$\gamma(\theta) = P(\text{Gamma}(2, \theta) \leq \log 4) = \begin{cases} 0.1534 & \theta = 2 \quad (\text{significance level } \alpha), \\ 0.4034 & \theta = 1 \quad (\text{power, since } H_0 \text{ is false}). \end{cases}$$

Thus the significance level is $\alpha = 0.1534$ and the power of the test (when H_0 is false, i.e. $\theta = 1$) is 0.4034.

8. Assume that the weight of cereal in a “10-ounce box” is $N(\mu, \sigma^2)$. To test $H_0 : \mu = 10.1$ against $H_1 : \mu > 10.1$, we take a random sample of size $n = 16$ and observe that $\bar{x} = 10.4$ and $s = 0.4$.

- (a) Do we accept or reject H_0 at the 5% significant level?
- (b) What is the approximate p-value of this test?

Sol.

- (a) We reject $H_0 : \mu = 10.1$ if

$$T = \frac{\bar{x} - 10.1}{s/\sqrt{n}} \geq t_{0.05, n-1} = t_{0.05, 15} = 1.754.$$

The value of T is

$$T = \frac{10.4 - 10.1}{0.4/\sqrt{16}} = 3.$$

Thus, we reject H_0 and conclude $\mu > 10.1$.

- (b) The approximate p -value is

$$P(t_{15} \geq 3) = 0.0045.$$